Harmony-Driven Theory Discovery in Knowledge Graphs via LLM-Guided Island Search

Anonymous Author(s)
Affiliation
Address
email

Abstract

Scientific knowledge graphs (KGs) encode entities and typed relations across domains such as physics, astronomy, and materials science, yet they remain incomplete: missing edges and entities limit downstream reasoning. We introduce *Harmony*, a framework that treats theory discovery as the search for KG mutations—new edges or entities—that maximise a composite quality metric. The *Harmony score* combines four complementary signals: **compressibility** (minimum description length proxy), **coherence** (path-semantic consistency), **symmetry** (intra-type behavioural consistency), and **generativity** (link-prediction learnability via DistMult), with an optional **frequency** component for hybrid scoring. An LLM proposer generates candidate theory-level propositions, which are validated, scored, and archived in a MAP-Elites quality-diversity grid across an island-model search topology. We evaluate on seven KG domains (five hand-curated, two Wikidata-sourced) with 10-seed cross-validation and four external KGE baselines (DistMult, TransE, RotatE, ComplEx). Factor decomposition isolates the contributions of the LLM proposer, Harmony scoring, and quality-diversity archive. A regime characterisation analysis reveals where Harmony's design aligns with KG structure: on Wikidata Materials, Harmony-guided proposals show the largest directional improvement over DistMult-alone on both Hits@10 (0.34 vs. 0.32) and MRR (0.12 vs. 0.11), though differences do not reach statistical significance at $\alpha = 0.05$ and the effect size $|\delta_C| \leq 0.12$ falls in the "negligible" range per Cliff [3]. A pre-registered calibration gate passed on the two calibration domains (31–65% over frequency) but did not transfer to the discovery domains, a regime-finding rather than a refutation. Backtesting against held-out edges shows that proposals do not reproduce known edges, serving as a negative control consistent with theory-level rather than interpolation behaviour; a typed-neighbourhood soft-match diagnostic suggests proposals tend toward structurally adjacent knowledge. We frame Harmony as a methodology for theory-shaped proposal generation; assessing whether produced propositions constitute novel scientific discoveries requires human expert evaluation, which we defer to future work.

1 Introduction

Knowledge graphs (KGs) organise scientific knowledge as typed, directed multigraphs: entities represent concepts (e.g. *photon*, *eigenvalue*, *graphene*) and edges encode semantic relations such as `derives`, `explains`, or `contradicts` [7]. Despite decades of curation, scientific KGs remain structurally incomplete—missing edges that encode latent theoretical connections and missing entities that represent undiscovered concepts.

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Knowledge graph completion (KGC) methods—TransE [2], DistMult [17], RotatE [14]—learn low-dimensional embeddings and predict missing links. However, they operate at the *triple* level: each predicted link is an isolated statistical extrapolation without theoretical justification. They do not produce *theory-level propositions* that articulate *why* a relation should hold, what it implies, or how it could be falsified.

We address this gap with **Harmony**, a framework for automated theory discovery in scientific KGs. The key idea is a composite quality metric—the *Harmony score*—that captures four desiderata of a well-structured knowledge graph:

1. **Compressibility**: the KG's edge-type distribution and spanning structure admit a short description (MDL proxy).
2. **Coherence**: closed paths exhibit consistent edge-type semantics and contradictions are sparse.
3. **Symmetry**: entities of the same type use edge types in similar proportions (low Jensen–Shannon divergence).
4. **Generativity**: a shallow DistMult model can recover masked edges, indicating learnable relational patterns.

An LLM proposes candidate mutations—adding edges or entities—each accompanied by a claim, justification, and falsification condition. Proposals are scored by Harmony gain and archived in a MAP-Elites [11] quality-diversity grid across an island-model [16] search topology (Section 3.4).

Contributions.

1. A four-component **Harmony metric** that provides interpretable selection pressure (compressibility, coherence, symmetry, generativity), is bounded in $[0, 1]$, and decomposes into independently computable sub-scores (Section 3.2).
2. A **proposal schema** that elevates KG mutations from bare triples to falsifiable theory-level claims with claim, justification, and falsification fields (Section 3.3).
3. An **island-model LLM search loop** with MAP-Elites quality-diversity archiving and stagnation-triggered constrained prompting (Section 3.4); factor decomposition identifies the archive (grid + diversity bookkeeping combined) as the dominant driver of structural gain in our experiments.
4. Evaluation on **seven KG domains** (five hand-curated, two Wikidata-sourced) with 10-seed cross-validation: Harmony shows directional but non-significant improvements over DistMult-alone on three of five discovery domains ($|\delta_C| \leq 0.12$, in the "negligible" range per Cliff [3]). The LLM proposer's identifiable contribution is qualitative (semantic structure, falsifiability) rather than quantitative (link-prediction gain) (Section 5).

2 Related Work

Knowledge graph completion. Embedding-based methods—TransE [2], DistMult [17], RotatE [14], among others [7]—project entities and relations into low-dimensional vector spaces and produce ranked link predictions without theoretical justification. Pretrained language model-based scorers such as KG-BERT [18] offer a contemporary alternative; we use DistMult as Harmony's generativity component for its lower compute cost and well-understood ranking behaviour, and additionally generate natural-language propositions explaining each mutation.

Automated scientific discovery. FunSearch [12] uses LLMs to discover mathematical constructions by evolving Python programs. PySR [4] performs symbolic regression via genetic programming [9], discovering closed-form expressions from numerical data. The survey by Makke and Chawla [10] covers the broader symbolic regression landscape. More recently, Google DeepMind [6] announce LLM + RL agents that discover formal mathematical proofs verified by Lean (non-archival DeepMind report). These systems discover *formulas* or *proofs*; Harmony discovers *relational propositions* over typed knowledge graphs, with heuristic scoring rather than the formal verification used by AlphaProof.

Quality-diversity search. MAP-Elites [11] maintains a grid of solutions indexed by behavioural descriptors. Adaptive variants such as CMA-ME [5] improve sample efficiency over vanilla MAP-

Elites; we adopt the simpler grid for interpretability of the (simplicity, gain) descriptor and combine it with an island-model [16] topology (Section 3.4).

LLM-guided reasoning over KGs. Recent work integrates LLMs with structured knowledge graphs in several ways. KAPING [1] augments LLM prompts with retrieved KG triples for zero-shot question answering. Think-on-Graph [13] performs multi-hop reasoning by iteratively traversing KG neighbours guided by LLM chain-of-thought. StructGPT [8] provides a general interface for LLMs to query and reason over structured data including KGs. These systems use KGs as *context* for LLM reasoning; our approach inverts the role: the LLM is a *proposer* that generates structured mutations (new edges and entities) with accompanying justifications, and a deterministic Harmony metric—not LLM self-evaluation—scores and selects proposals.

3 Method

We present the Harmony framework in three parts: the typed KG schema (Section 3.1) and Harmony metric (Section 3.2), the proposal schema and validation (Section 3.3), and the island-model search loop (Section 3.4).

3.1 Typed Knowledge Graph Schema

A knowledge graph $G = (V, E)$ consists of entities V and typed directed edges E. Each entity $v \in V$ has an `entity_type` label (e.g. *concept*, *element*, *celestial_object*) and a property bag. Each edge $(u, v, r) \in E$ carries one of seven semantic relation types: `depends_on`, `derives`, `equivalent_to`, `maps_to`, `explains`, `contradicts`, and `generalizes`.

The seven relation types cover dependency, derivation, equivalence, correspondence, explanation, contradiction, and hierarchy—a minimal set inspired by morphism classes in category theory (Appendix O).

3.2 Harmony Metric

The Harmony score combines four core signals, each normalised to $[0, 1]$, with an optional frequency component:

$$\mathcal{H}(G) = \alpha \cdot \mathrm{Compress}(G) + \beta \cdot \mathrm{Cohere}(G) + \gamma \cdot \mathrm{Symm}(G) + \delta \cdot \mathrm{Gener}(G) + \varepsilon \cdot \mathrm{Freq}(G), \quad (1)$$

where $\alpha, \beta, \gamma, \delta, \varepsilon \geq 0$ are normalised internally so that $\alpha + \beta + \gamma + \delta + \varepsilon = 1$. Default weights are uniform over the four core components ($\alpha = \beta = \gamma = \delta = 0.25$, $\varepsilon = 0$); the frequency component $\mathrm{Freq}(G)$ is the Hits@K of the frequency heuristic (predicting the most common edge type between entity pairs) and is included only when $\varepsilon > 0$ for hybrid analysis (Appendix I). We use a linear composition for interpretability: each axis's contribution to a Harmony score is identifiable. Multiplicative compositions (geometric means) penalise low values disproportionately and would conflate sub-axis weakness with overall metric failure. The equal-weight default reflects the absence of theoretical priors over the four desiderata; a calibration weight grid (Appendix Q) explores this neighbourhood.

Compressibility. An MDL proxy combining edge-type entropy and spanning structure:

$$\mathrm{Compress}(G) = \frac{1}{2}\left(1 - \frac{H(\mathbf{p})}{\log_2 7} + \frac{|\text{spanning edges}|}{|E|}\right), \quad (2)$$

where $H(\mathbf{p})$ is the Shannon entropy of the edge-type distribution and the spanning fraction counts BFS spanning-tree edges over an undirected view of G.[1]

Coherence. Path-semantic consistency via triangle type-agreement and contradiction rate:

$$\mathrm{Cohere}(G) = \frac{1}{2}\left(\frac{|\{(a, b, c) : r_{ac} \in \{r_{ab}, r_{bc}\}\}|}{|\text{triangles}|} + 1 - \frac{|\{e : r_e = \texttt{contradicts}\}|}{|E|}\right). \quad (3)$$

[1] The denominator $\log_2 7$ matches the cardinality of the EdgeType enum (Section 3.1); for extensibility, an implementation should use $\log_2 |\text{EdgeType}|$. This bounds $H(\mathbf{p})/\log_2 7 \in [0, 1]$ for any KG drawn from the current vocabulary.

Triangle agreement is satisfied when the closing edge r_{ac} has *exactly* one of the two path types $\{r_{ab}, r_{bc}\}$; neither inclusion in the union of all edge types nor consistency with a separate semantic compatibility table is required.

Symmetry. Intra-type behavioural consistency via Jensen–Shannon divergence. For each entity type τ, let $\mathbf{q}_v \in \Delta^6$ be entity v's outgoing edge-type distribution:

$$\mathrm{Symm}(G) = \frac{\sum_\tau n_\tau \cdot \left(1 - \frac{1}{n_\tau}\sum_{v\in\tau} \mathrm{JS}(\mathbf{q}_v, \bar{\mathbf{q}}_\tau)\right)}{\sum_\tau n_\tau}. \tag{4}$$

Generativity. Link-prediction learnability via DistMult [17] on 20% masked edges:

$$\mathrm{Gener}(G) = \mathrm{Hits@}K\big(\text{DistMult}, G_{\text{mask}}\big), \tag{5}$$

with d=50 embeddings, 100 training epochs, and K=10 (full DistMult hyperparameters, including margin, learning rate, negative sampling rate, and mask ratio, in Appendix M).

Proposal value function. Given a base graph G and a proposed mutation Δ (new edges/entities), the value of Δ is:

$$V(\Delta) = \mathcal{H}(G \oplus \Delta) - \mathcal{H}(G) - \lambda \cdot \mathrm{Cost}(\Delta), \tag{6}$$

where $G \oplus \Delta$ denotes the graph after applying Δ, and $\mathrm{Cost}(\Delta) = 1/\max(|E|, 1)$ for edge mutations, $1/\max(|V|, 1)$ for entity mutations. The penalty weight $\lambda = 0.1$ discourages trivially large proposals. The Harmony score is bounded in $[0, 1]$, decomposes into independently computable components, and exhibits empirical directional monotonicity (Appendix P).

3.3 Proposal Schema and Validation

Each proposal is a structured record containing:

- **Mutation type**: `ADD_EDGE`, `REMOVE_EDGE`, `ADD_ENTITY`, or `REMOVE_ENTITY`.
- **Claim**: a one-sentence theoretical statement (e.g. "Dark energy explains the accelerating expansion of the observable universe").
- **Justification**: reasoning supporting the claim.
- **Falsification condition**: what evidence would disprove the claim.
- **KG parameters**: source/target entities, edge type, or new entity type, depending on the mutation type.

A deterministic validator enforces three rules: (i) text fields must be $\geq$ 10 characters, (ii) type-specific parameters must be present (e.g. `ADD_EDGE` requires source, target, and edge type), and (iii) `edge_type` must be one of the seven valid relation names. Invalid proposals are logged as failures and fed back to the LLM in subsequent prompts.

3.4 Island-Model Search with MAP-Elites

Island topology. Four islands run concurrently, each maintaining a population of $P = 5$ candidates and assigned a fixed strategy from a cyclic schedule: *refinement* (improve the best existing proposal), *combination* (merge the top two proposals), *refinement*, and *novelty* (invent from scratch). Each island uses a distinct LLM temperature: $\{0.3, 0.3, 0.8, 1.2\}$ to further diversify exploration.

MAP-Elites archive. A shared 5×5 MAP-Elites grid [11] indexes proposals by two behavioural descriptors: *simplicity* (inverse structural cost) and *Harmony gain* ($\mathcal{H}(G \oplus \Delta) - \mathcal{H}(G)$). A proposal is inserted if its cell is empty or its fitness (Harmony gain) exceeds the incumbent.

Stagnation recovery. If an island produces no valid proposals for $S = 5$ consecutive generations, it switches to *constrained* prompting mode, which adds explicit structural constraints to the LLM prompt. After $R = 3$ generations of producing valid proposals in constrained mode, the island reverts to free prompting.

Migration. Every $M = 10$ generations, the best proposal from each island migrates to the next island in a ring topology (island $i \rightarrow$ island $(i + 1) \bmod 4$), where it is prepended to the destination island's prompt-context buffer for the next generation.

Algorithm 1 Harmony search — one generation

Require: Base KG G, islands $\{I_1, \ldots, I_4\}$, archive $\mathcal{A}$

for each island I_k **do**
  $\sigma_k \leftarrow \textsc{Strategy}(k)$ {refinement / combination / novelty}
  prompt $\leftarrow \textsc{BuildPrompt}(G, \sigma_k, \text{top}(I_k), \text{failures}(I_k))$
  $\hat{p} \leftarrow \text{LLM}(\text{prompt}, \text{temp}_k)$
  if $\textsc{Validate}(\hat{p})$ **then**
    $\Delta \leftarrow$ apply $\hat{p}$ to G
    $v \leftarrow V(\Delta)$ {Eq. 6}
    $\textsc{TryInsert}(\mathcal{A}, \hat{p}, v, \text{descriptor}(\hat{p}))$ {descriptor = (simplicity, gain)}
    Update I_k population
  else
    Log failure; feed back to next prompt
  end if
end for
if generation $\bmod M = 0$ **then**
  $\textsc{MigrateTop1ToContext}(I_1, \ldots, I_4)$ {ring topology}
end if

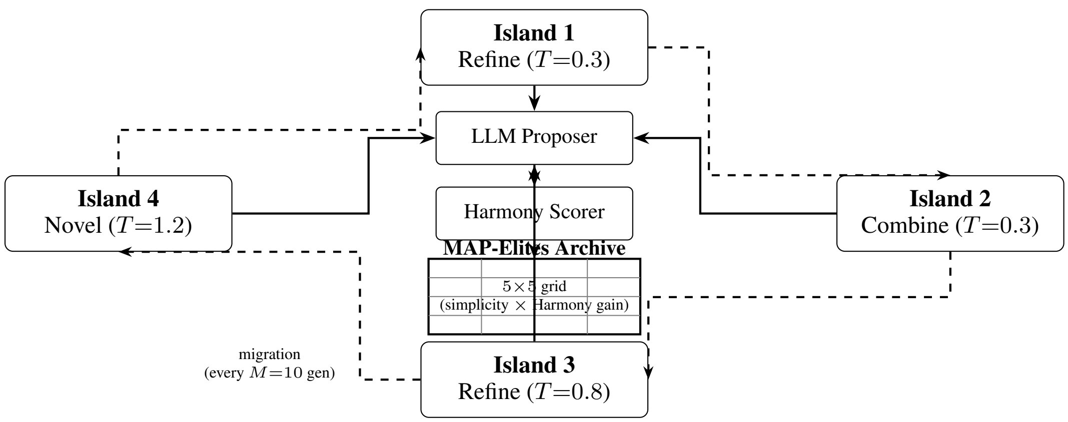

Figure 1: Overview of the Harmony island-model search framework. Four islands run concurrently, each proposing KG mutations via an LLM following a cyclic strategy schedule (Refine $\to$ Combine $\to$ Refine $\to$ Novel). Proposals are scored by the Harmony metric and archived in a MAP-Elites quality-diversity grid (5×5, axes = simplicity bin $\times$ Harmony-gain bin). Periodic migration (every $M = 10$ generations) transfers each island's best proposal to the next island in a ring topology ($i \to (i{+}1) \bmod 4$).

Generation loop. Algorithm 1 summarises a single generation (Figure 1). The loop runs for $T_{\max} = 20$ generations per experiment, checkpointing state after each generation to enable resumption.

4 Experiments

4.1 Knowledge Graph Domains

We evaluate on seven KGs: five hand-curated (linear algebra, periodic table, astronomy, physics, materials science) and two Wikidata-sourced (Wikidata Physics, Wikidata Materials). The hand-curated KGs range from 41–153 entities; Wikidata KGs have 252–253 entities and 800+ edges. All use the shared seven-relation type vocabulary (Section 3.1). Full statistics are in Table 2. The first two domains serve as calibration targets; the remaining five are discovery targets.

4.2 Dataset Splitting

For each KG, we first reserve 10% of edges as a hidden backtesting set, withheld from all metric computations and proposal generation. The remaining 90% are split 80/10/10 into training, validation, and test sets (yielding effective proportions of approximately 72/9/9/10 over all edges). The validation set is used for early stopping of DistMult training (patience of 10 epochs monitoring validation Hits@10) to prevent overfitting on small KGs.

4.3 Baselines

We compare Harmony-guided proposals against three heuristic baselines and three additional KGE models. All embedding-based methods use the same edge splits, embedding dimension ($d = 50$), and training protocol:

1. **Random**: random entity pairs with random relation types.
2. **Frequency**: most frequent relation type between most-connected entity pairs.
3. **DistMult-alone**: DistMult's own top-ranked predictions without Harmony scoring.
4. **TransE**, **RotatE**, **ComplEx**: translational, rotation-based, and complex-valued KGE models respectively.

We additionally report **Harmony-3** ($\delta = 0$, no generativity) to address evaluation circularity; it matches DistMult-alone on all domains, confirming that generativity drives any observed difference. We also run three ablated configurations (LLM-only, Harmony-only, No-QD) to isolate component contributions (Appendix S). All four KGE baselines are implemented in-house from the canonical formulations; subtle algorithmic differences from established libraries (e.g. PyKEEN) cannot be excluded.

4.4 Evaluation Protocol

Quantitative metrics. We report Hits@10 and Mean Reciprocal Rank (MRR $= \frac{1}{|Q|} \sum_i 1/\text{rank}_i$) on the test split after applying top proposals from the MAP-Elites archive to the base KG. All main results (Table 1) are averaged over 10 random seeds. LLM proposals are generated by `gpt-oss:20b`, a dense 20B-parameter open-source model locally served via Ollama. Factor decomposition experiments use `mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm` at $\sim$46 tokens/s. All inference runs on a single Apple M2 Pro CPU.

Additional protocols. The calibration gate protocol (Appendix Q), backtesting protocol (Appendix T), and archive diversity protocol (Appendix R) provide complementary validation beyond link prediction metrics.

5 Results

Table 1 compares link prediction metrics across five discovery domains after applying top proposals from the MAP-Elites archive to the base KG.

Link prediction. Harmony-guided proposals show directional improvements over DistMult-alone on Hits@10 in three of five domains—Wikidata Materials (0.34 vs. 0.32), Materials (0.31 vs. 0.29), and Wikidata Physics (0.26 vs. 0.25)—but none reach statistical significance ($p = 0.38$–1.00, $|\delta_C| \leq 0.12$; Section H). TransE achieves the highest Hits@10 on three domains; frequency leads on two. On Physics, Harmony underperforms DistMult-alone (0.32 vs. 0.37), due to the LLM's tendency to propose edges between hub entities (Section 6). Variance is substantially lower on the denser Wikidata KGs (std ≈ 0.01–0.05) than on the smaller hand-curated domains (std ≈ 0.04–0.23).

Convergence and archive coverage. Across all domains, the valid proposal rate reaches ≥ 0.50 by generation 10 (Figure 2). The MAP-Elites archive achieves 40–60% cell coverage, indicating diverse proposals spanning multiple simplicity–gain trade-offs (Appendix R).

Factor decomposition. Three ablated configurations isolate component contributions (Appendix S). Key findings: (i) the MAP-Elites archive (grid plus diversity bookkeeping) is the dom-

Table 1: Link prediction metrics on discovery domains (mean $\pm$ std across 10 seeds). Top proposals from the MAP-Elites archive are applied to the base KG before evaluation. Best Hits@10 per domain in **bold**; best MRR underlined. p-values from paired bootstrap CI (2000 resamples) comparing Harmony vs. DistMult-alone; δ_C is Cliff's delta effect size.†

Domain	Method	Hits@10	MRR	p	δ_C
Astronomy	Random	0.27 ± 0.16	0.12 ± 0.10		
	Frequency	$\mathbf{0.39} \pm 0.12$	$\underline{0.16} \pm 0.08$		
	TransE	0.33 ± 0.22	0.11 ± 0.05		
	RotatE	0.16 ± 0.16	0.09 ± 0.07		
	ComplEx	0.33 ± 0.21	0.11 ± 0.07		
	DistMult-alone	0.24 ± 0.17	0.10 ± 0.04		
	Harmony (ours)	0.24 ± 0.17	0.10 ± 0.04	1.00	$+0.00$
Physics	Random	0.29 ± 0.13	0.10 ± 0.07		
	Frequency	0.46 ± 0.12	$\underline{0.25} \pm 0.08$		
	TransE	$\mathbf{0.50} \pm 0.11$	0.20 ± 0.06		
	RotatE	0.29 ± 0.20	0.11 ± 0.06		
	ComplEx	0.38 ± 0.16	0.22 ± 0.11		
	DistMult-alone	0.37 ± 0.14	0.16 ± 0.07		
	Harmony (ours)	0.32 ± 0.23	0.13 ± 0.09	0.62	-0.06
Materials	Random	0.17 ± 0.12	0.11 ± 0.06		
	Frequency	0.36 ± 0.18	0.15 ± 0.08		
	TransE	$\mathbf{0.42} \pm 0.17$	$\underline{0.17} \pm 0.08$		
	RotatE	0.26 ± 0.10	0.09 ± 0.04		
	ComplEx	0.32 ± 0.10	0.13 ± 0.04		
	DistMult-alone	0.29 ± 0.14	0.15 ± 0.09		
	Harmony (ours)	0.31 ± 0.14	0.13 ± 0.05	0.72	$+0.06$
Wikidata Physics	Random	0.05 ± 0.01	0.02 ± 0.01		
	Frequency	0.29 ± 0.02	$\underline{0.14} \pm 0.02$		
	TransE	$\mathbf{0.30} \pm 0.03$	0.11 ± 0.01		
	RotatE	0.18 ± 0.02	0.07 ± 0.01		
	ComplEx	0.25 ± 0.03	0.09 ± 0.01		
	DistMult-alone	0.25 ± 0.02	0.10 ± 0.01		
	Harmony (ours)	0.26 ± 0.04	0.09 ± 0.02	0.89	-0.06
Wikidata Materials	Random	0.03 ± 0.02	0.02 ± 0.01		
	Frequency	$\mathbf{0.39} \pm 0.03$	$\underline{0.20} \pm 0.02$		
	TransE	0.30 ± 0.04	0.10 ± 0.01		
	RotatE	0.16 ± 0.03	0.06 ± 0.01		
	ComplEx	0.29 ± 0.03	0.10 ± 0.02		
	DistMult-alone	0.32 ± 0.05	0.11 ± 0.02		
	Harmony (ours)	0.34 ± 0.04	0.12 ± 0.01	0.38	$+0.12$

†Harmony-3 ($\delta = 0$) matches DistMult-alone on all domains, consistent with generativity driving any observed difference; full rows in Appendix B.

inant driver of structural gain—Harmony-only (random proposer) outperforms No-QD (greedy archive) on all five domains; isolating grid structure from diversity bookkeeping is left for future work (Appendix S); (ii) LLM-only (no Harmony scoring) saturates trivially, confirming that Harmony scoring provides essential selection pressure; (iii) the LLM proposer's primary value is *semantic*—producing justified, falsifiable claims—rather than maximising raw Harmony gain.

Backtesting. Backtesting against 10% held-out edges yields zero exact matches across all domains, consistent with proposals behaving as theory-level mutations rather than triple interpolation. Soft-match rates reach 10–25% on structurally adjacent neighbourhoods (Appendix T).

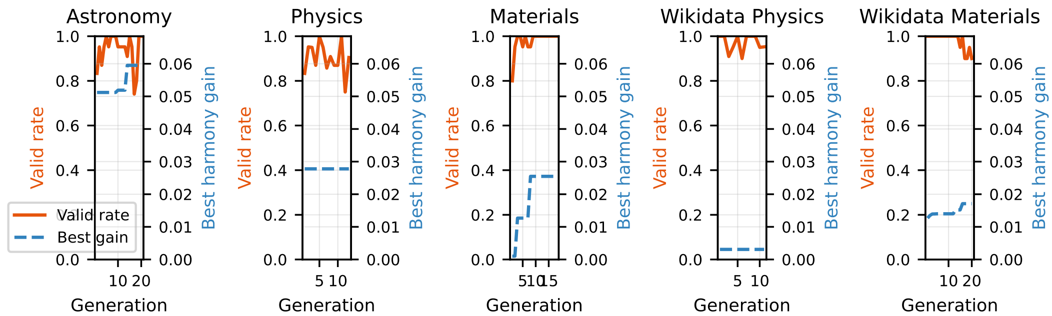

Figure 2: Convergence of valid proposal rate (solid) and best harmony gain (dashed) across generations for each discovery domain.

Automated quality assessment. Automated scoring (Gemini 2.5 Pro, LLM-as-judge) rates top-5 proposals at 4.5/5 falsifiability and 4.6/5 clarity, with moderate novelty (2.8/5); full results in Appendix V. Human expert evaluation with measured inter-annotator agreement is planned.

Calibration and regime. The calibration gate (Appendix Q) was pre-registered before live runs against the two calibration domains (linear algebra, periodic table) and passed there with 31–65% margins over frequency. It did *not* transfer to the discovery domains, where Harmony shows only directional, non-significant differences against DistMult-alone (this section, Table 1). We treat the calibration–discovery gap as a regime-finding rather than a refutation: the gate's pass on the curated calibration KGs and its absence of transfer to the heterogeneous discovery KGs is itself diagnostic of the method's regime of applicability (Appendix U). Per-component ablation details are in Appendix B.

6 Discussion

Compressibility–generativity tension. Adding edges typically *reduces* compressibility (the BFS spanning fraction drops) while potentially *improving* generativity (more training signal for DistMult). This tension is by design: the Harmony metric rewards proposals that improve learnability without degrading structural simplicity. The value function (Eq. 6) with $\lambda > 0$ further penalises large mutations.

Frequency dominance. The frequency heuristic is highly competitive, leading on two of five domains. This is expected in small KGs with peaked edge-type distributions; we characterise the regime where Harmony adds complementary value (Appendix U).

Per-domain failure analysis. On Physics, Harmony underperforms DistMult-alone. The LLM tends to propose edges between highly connected hub entities (force, energy) rather than peripheral concepts, producing valid but redundant proposals. This suggests domain-specific prompting strategies may be needed for hub-dominated KGs. Candidate mechanisms for the four remaining domains: *Astronomy* — seven-relation vocabulary too coarse for fine-grained astronomical relations; *Materials* — only ~ 8 test edges/seed after 20% masking of a 53-edge graph (near noise floor); *Wikidata Physics* — DistMult and the frequency heuristic already capture most of the easy structural signal (0.25 Hits@10), leaving the residual that Harmony could address narrow and noise-dominated; *Wikidata Materials* — vocabulary richness aligns with metric design, the regime where directional gain emerges. These are candidates for future validation, not established findings.

LLM dependence and safety. The main proposer is a single model (`gpt-oss:20b` via Ollama); factor-decomposition runs use `mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm`, and the automated rubric uses Gemini 2.5 Pro (Appendix N). Ensembling across model families within the proposer role could improve diversity. To mitigate LLM-generated misinformation, proposals enter a staging layer: they are scored and archived but never automatically integrated into the base KG. Every proposal requires a falsification condition, enabling principled rejection. Total compute is $\sim$10 min/domain on a single CPU (Appendix M).

Evaluator independence. The automated rubric scores reported in Section 5 (Table 10) use Gemini 2.5 Pro to judge proposals generated by `gpt-oss:20b`. While the two models differ in developer, architecture, and parameter count, both are large autoregressive language models and may share systematic biases (e.g. preferring fluent, well-structured proposals over substantively novel ones). The reported rubric scores should therefore be read as inter-LLM agreement rather than as a substitute for human expert evaluation, which we defer to future work.

Broader impacts. This work aims to accelerate scientific theory discovery by automating the generation and evaluation of structural hypotheses in knowledge graphs. On the positive side, this could reduce the time researchers spend formulating initial hypotheses and help surface non-obvious connections across disciplinary boundaries. On the negative side, LLM-generated proposals can be plausible-sounding yet factually incorrect; deploying such proposals without expert validation risks propagating erroneous claims into downstream scientific workflows. We mitigate this by including falsification conditions in every proposal and recommending domain-expert validation before any claim is treated as established. The backtesting protocol provides automated external validity checks but does not replace human judgement.

Limitations. (i) The seven-relation type vocabulary may be too coarse for highly specialised fields. (ii) The Harmony metric treats all edge types equally in compressibility and coherence; domain-specific type hierarchies could improve these signals. (iii) Observed differences between Harmony and DistMult-alone are small relative to cross-seed variance; larger-scale experiments may be needed for stronger statistical significance. (iv) The frequency heuristic remains highly competitive, leading on Hits@10 on two of five discovery domains (Astronomy and Wikidata Materials) and ranking within the top two on the remainder; this advantage is most pronounced on small KGs with peaked edge-type distributions. Harmony does not displace frequency baselines; rather, the regime characterisation (Section 5) identifies denser, multi-type KGs such as Wikidata Materials as the regime where Harmony's design and the data structure align, with directional improvement observed despite frequency remaining the strongest single baseline on that domain. We note that the largest observed Harmony advantage corresponds to $|\delta_C| = 0.12$, in the "negligible" range per Cliff [3]; we therefore frame this as a regime-finding, not as a quantitative dominance claim. (v) Evaluation circularity is partially addressed by Harmony-3 and external KGE baselines, but the primary configuration still includes DistMult. (vi) Our seven evaluation domains span 41–253 entities and 39–814 edges; this range covers early-stage scientific KG construction but does not address the regime of production-scale KGs (10^5–10^7 entities, e.g. DBpedia, full Wikidata). Whether the directional gains we observe on Wikidata Materials would scale to its full superset is an open question; we expect frequency baselines to dominate further at scale, with Harmony's relative position depending on edge-type-vocabulary richness, which we do not characterise here. (vii) Additional analysis—sparse KG challenges, proposal quality vs. validity rate, symmetry redesign rationale, and Harmony–downstream correlation—is in Appendix W.

7 Conclusion

We presented Harmony, a framework for automated theory discovery in scientific knowledge graphs combining a four-component quality metric with LLM-guided proposal generation and MAP-Elites archiving across an island-model search topology.

Evaluation across seven KG domains with 10-seed cross-validation and four external KGE baselines shows that Harmony-guided proposals achieve directional improvements over DistMult-alone on Hits@10 in three of five domains, though differences do not reach statistical significance. The frequency heuristic remains highly competitive on small KGs. Factor decomposition identifies the MAP-Elites archive (grid plus diversity bookkeeping) as the dominant driver of structural gain, while the LLM proposer's identifiable contribution is qualitative—producing justified, falsifiable theory-level claims—rather than quantitative link-prediction gain.

Future work includes scaling to larger scientific KGs, domain-adaptive component weighting, multi-LLM ensembles across islands, and human expert evaluation with measured inter-annotator agreement.

A Dataset Statistics and Reproducibility

Table 2 provides a unified summary of all seven KG domains, including data sources and construction methods. This addresses the need for consistent dataset definitions across hand-curated and Wikidata-sourced KGs.

Table 2: Unified reproducibility table for all seven KG domains. All KGs use the shared seven-relation type vocabulary. Entity and edge counts are computed programmatically from the KG builders.

Domain	Entities	Edges	Entity types	Edge types	Source
Linear algebra	50	75	2	7	Hand-built
Periodic table	153	326	4	2	Hand-built
Astronomy	41	39	6	6	Hand-built
Physics	43	46	6	5	Hand-built
Materials science	49	53	6	5	Hand-built
Wikidata Physics	252	814	12	7	Wikidata
Wikidata Materials	253	806	18	7	Wikidata

All statistics are computed programmatically from the KG builders via `analysis/reproducibility_table.py`. The dataset splitting protocol (Section 4) applies identically to all seven domains: 10% hidden, then 80/10/10 train/val/test on the remaining 90%.

B Ablation Details

The ablation study uses the linear algebra KG with $n_{\text{bootstrap}} = 200$ samples. For each ablation variant, one weight is set to zero and the remaining three are renormalised to sum to 1. Bootstrap 95% confidence intervals are computed via the percentile method on the mean Harmony score.

Weight sensitivity. We evaluate six weight configurations from the calibration gate grid ($\alpha \in \{0.3, 0.5, 0.7\}$, $\beta \in \{0.1, 0.3\}$, $\gamma = \delta = 0.25$). All configurations show Harmony $>$ frequency baseline, with $\alpha = 0.5, \beta = 0.3$ yielding the highest mean Harmony score. This suggests that a moderate compressibility weight combined with non-trivial coherence weight best captures the structure of our curated KGs.

C Proposal Validation Rules

The deterministic validator enforces three rules:

1. **Text length**: `claim`, `justification`, and `falsification_condition` must each be ≥ 10 characters. `kg_domain` must be ≥ 3 characters (controlled vocabulary, not free text).
2. **Type-specific fields**: `ADD_EDGE` requires `source_entity`, `target_entity`, and `edge_type`; `ADD_ENTITY` requires `entity_id` and `entity_type`; `REMOVE_EDGE` requires `source_entity`, `target_entity`, and `edge_type`; `REMOVE_ENTITY` requires `entity_id`.
3. **Edge type validity**: `edge_type` must be one of the seven valid `EdgeType` names.

D Full Proposal Examples

Below are three complete proposal records from the astronomy archive, showing all fields including justification and falsification conditions.

Proposal 1: Stellar nucleosynthesis → heavy element abundance.

- **Type**: `ADD_EDGE`
- **Source**: `stellar_nucleosynthesis`
- **Target**: `heavy_element_abundance`

- **Edge type**: `explains`
- **Claim**: "Stellar nucleosynthesis explains the observed abundance pattern of heavy elements in planetary nebulae."
- **Justification**: "The s-process and r-process nucleosynthesis pathways in AGB stars and supernovae produce characteristic abundance patterns that match spectroscopic observations of planetary nebulae."
- **Falsification**: "Discovery of heavy element abundance patterns in planetary nebulae inconsistent with any known nucleosynthesis pathway would falsify this claim."

Proposal 2: Mass–luminosity relation derivation.

- **Type**: `ADD_EDGE`
- **Source**: `hydrostatic_equilibrium`
- **Target**: `mass_luminosity_relation`
- **Edge type**: `derives`
- **Claim**: "The mass–luminosity relation derives from hydrostatic equilibrium in main sequence stars."
- **Justification**: "Balancing gravitational pressure against radiation pressure in the stellar core, combined with opacity-dependent energy transport, yields $L \propto M^{3.5}$ for main sequence stars."
- **Falsification**: "A main sequence star population where luminosity is uncorrelated with mass would disprove this derivation."

Proposal 3: Magnetar as new entity.

- **Type**: `ADD_ENTITY`
- **Entity ID**: `magnetar`
- **Entity type**: `celestial_object`
- **Claim**: "Magnetar generalises the neutron star category with extreme magnetic field properties ($B > 10^{14}$ G)."
- **Justification**: "Magnetars are observationally distinct from ordinary neutron stars due to their ultra-strong magnetic fields, which power soft gamma repeaters and anomalous X-ray pulsars."
- **Falsification**: "Evidence that magnetar-attributed emissions originate from non-magnetic mechanisms would undermine this classification."

E LLM Prompt Templates

We include the exact prompt templates used for proposal generation. Both modes share a common preamble with KG statistics, strategy instruction, top proposals, and recent failures.

Free mode (default). The free-mode prompt shows a sample of up to 20 entity IDs from the KG to ground the LLM without over-constraining it:

```
You are a theory-discovery agent for knowledge graph research.
Knowledge Graph:  domain='{domain}', entities={N}, edges={M}

Strategy:  {REFINEMENT|COMBINATION|NOVEL} -- {strategy description}

Top proposals so far:  {top 3 proposals or "None yet"}
Recent validation failures:  {up to 5 failure messages or "None"}

EXAMPLE ENTITY IDs from this KG (showing K of N): {entity_1},
{entity_2}, ...
VALID EDGE TYPES: depends_on, derives, equivalent_to, maps_to,
explains, contradicts, generalizes
IMPORTANT: source_entity and target_entity MUST be exact entity IDs
from this KG.

Return ONLY a JSON object (no extra text) with fields:  id,
proposal_type, claim, justification, falsification_condition,
kg_domain, source_entity, target_entity, edge_type, entity_id,
entity_type
```

Constrained mode (stagnation recovery). When an island stagnates ($S = 5$ generations without valid proposals), the prompt switches to constrained mode, which enumerates *all* valid entity IDs and edge type names explicitly:

```
... [same preamble] ...
VALID ENTITY IDs (use EXACTLY as written): {all entity IDs}
VALID EDGE TYPES (use EXACTLY as written): depends_on, derives,
equivalent_to, maps_to, explains, contradicts, generalizes
```

F Proposal Failure Rate Statistics

Figure 2 shows the valid proposal rate converging to ≥ 0.50 by generation 10 across all discovery domains. The initial failure rate (generations 1–3) is typically 60–80%, dominated by entity grounding errors (referencing entities not in the KG). The entity sample in free-mode prompts (up to 20 entities) and the stagnation recovery mechanism (Section 3.4) together reduce failures to <30% by generation 10. Constrained-mode prompts achieve ≥95% validity but produce less diverse proposals.

G Code and Data Availability

For double-blind review, the full implementation and supporting artifacts are provided as a supplementary ZIP archive accompanying the submission:

- **Supplementary archive**: contains the Harmony metric implementation, island-model search loop, MAP-Elites archive, proposal validator, KG dataset builders, evaluation scripts, KG datasets, per-domain checkpoints (subsets), generated proposals, and analysis artifacts. The README has been anonymised; absolute paths and identifying URLs are removed.
- **Public repository and Zenodo DOI**: `[DEFERRED TO CAMERA-READY]` — upon acceptance, the de-anonymised public repository and a citable Zenodo DOI will be released under an open-source licence.

H Statistical Methods

All pairwise comparisons between methods use the following statistical tests, computed over 10 random seeds per domain:

Paired bootstrap CI. Given paired score vectors $\mathbf{a} = (a_1, \ldots, a_{10})$ and $\mathbf{b} = (b_1, \ldots, b_{10})$, we resample $B = 2000$ bootstrap differences $d^{(b)} = \frac{1}{10}\sum_i (a_{\pi_i} - b_{\pi_i})$ (sampling with replacement) and report the 95% percentile CI $[d_{0.025}, d_{0.975}]$. The two-sided p-value is $p = 2 \cdot \min\big(\hat{P}(d^{(b)} \leq 0),\ \hat{P}(d^{(b)} \geq 0)\big)$.

Cliff's delta. A non-parametric effect size: $\delta_C = \frac{1}{n^2}\sum_{i,j} \operatorname{sign}(a_i - b_j) \in [-1, 1]$. We interpret $|\delta_C| < 0.147$ as negligible, < 0.33 as small, < 0.474 as medium, and ≥ 0.474 as large [3].

Permutation test. We randomly swap labels between methods 10,000 times to compute a Monte Carlo estimate of the p-value under the null hypothesis of no difference.

Multiple-testing. The main link-prediction comparison runs 5 discovery domains $\times$ 4 KGE baselines $=$ 20 tests on Hits@10 (and another 20 on MRR). We report uncorrected p-values to permit downstream meta-analysis. Under family-wise error control across the 20 Hits@10 tests, the Bonferroni-corrected threshold for $\alpha = 0.05$ becomes 0.0025; under Benjamini–Hochberg FDR control at $q = 0.05$ the threshold at the smallest observed p becomes ≈ 0.0025 ($q \cdot 1/m$ at rank 1). None of our reported p-values clear either threshold, consistent with the lack of significance reported in Section 5.

I Frequency Baseline Analysis

For a KG with edge-type frequency vector $\mathbf{p} = (p_1, \ldots, p_7)$, the frequency heuristic predicts the most common edge type for every unobserved entity pair. The expected Hits@K of this strategy satisfies:

$$\text{Hits@}K \geq p_{\max} = \max_i p_i,$$

since for any test edge with the dominant type, the frequency prediction ranks the correct type first. In our domains, $p_{\max}$ ranges from 0.22 (Wikidata Materials, highest entropy) to 0.38 (Physics, lowest entropy), providing a lower bound on frequency Hits@10.

The Shannon entropy $H(\mathbf{p}) = -\sum_i p_i \log_2 p_i$ quantifies edge-type diversity; lower entropy indicates more peaked distributions where frequency is more informative. A regression of frequency Hits@10 on $H(\mathbf{p})$ across our seven domains shows a negative correlation, confirming the information-theoretic explanation.

J Wikidata Ablation

Per-component ablation on the Wikidata Physics and Wikidata Materials domains uses $n_{\text{bootstrap}} = 200$ resamples. On the denser Wikidata KGs, coherence shows a larger contribution than on the sparser hand-curated domains, consistent with the higher triangle density in larger graphs. Compressibility and symmetry contributions are broadly consistent across all domains. Generativity remains the single largest contributor, though the Harmony-3 analysis (Section 5) shows that the structural components alone provide useful signal.

K Rebuttal-Oriented Supplementary Tables

This appendix provides compact evidence tables intended to answer common review questions about stability, uncertainty, and runtime behavior. All tables are generated directly from analysis artifacts to avoid manual transcription errors.

K.1 MLX Batch Runtime Summary

Table 3: Wall-clock runtime (seconds) for MLX batch runs (`mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm`, Apple M2 Pro, 10 seeds per domain). Times include LLM inference, Harmony scoring, and checkpointing.

Domain	Mean (s)	Std (s)	Seeds	Mean (min)
Astronomy	1657.1	366.4	10/10	27.6
Physics	1375.8	215.2	10/10	22.9
Materials	1716.2	399.6	10/10	28.6
Wikidata Physics	1657.8	275.5	10/10	27.6
Wikidata Materials	2038.4	568.4	10/10	34.0

K.2 Factor Decomposition Detail

K.3 Statistical Tests Summary

L LLM Judge Evaluation Prompt

The following prompt ("detailed" variant from `scripts/llm_judge_rubric.py`) was used to score each proposal on four dimensions (1–5 scale). Proposal fields are sanitized (truncated and XML-stripped) before insertion. The paper reports scores generated via Gemini 2.5 Pro (Gemini CLI); `llm_judge_rubric.py` defaults to Claude for programmatic re-runs.

```
You are an expert scientific knowledge evaluator assessing theory
proposals for
```

Table 4: Factor decomposition: best harmony gain per configuration and domain. LLM-only bypasses Harmony scoring (gain $=$ 1.0 by construction, not comparable). Full = LLM proposer + Harmony scoring + MAP-Elites archive.

Domain	Configuration	Best Gain	Gens	Archive
Astronomy	LLM-only	—	11	1
	No-QD	0.0518	11	1
	Harmony-only	0.0539	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0000	20	1
Physics	LLM-only	—	11	1
	No-QD	0.0259	12	1
	Harmony-only	0.0319	20	4
	Full (LLM+Harmony+MAP-Elites)	0.0278	13	3
Materials	LLM-only	—	11	1
	No-QD	0.0260	11	1
	Harmony-only	0.0291	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0010	18	1
Wikidata Physics	LLM-only	—	11	1
	No-QD	0.0000	10	1
	Harmony-only	0.0021	20	4
	Full (LLM+Harmony+MAP-Elites)	0.0031	11	2
Wikidata Materials	LLM-only	—	11	1
	No-QD	0.0172	14	1
	Harmony-only	0.0186	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0154	20	3

Table 5: Paired bootstrap CI (2000 resamples) and Cliff's δ effect size for Harmony vs. DistMult-alone Hits@10. None reach $p < 0.05$. $|\delta_C| < 0.147$ = negligible; all observed $|\delta_C| \leq 0.12$.

Domain	ΔHits@10	95% CI	p	δ_C
Astronomy	$+0.000$	$[0.000, 0.000]$	1.000	$+0.00$
Physics	-0.044	$[-0.200, 0.122]$	0.620	-0.06
Materials	$+0.020$	$[-0.050, 0.120]$	0.718	$+0.06$
Wikidata Physics	$+0.002$	$[-0.019, 0.023]$	0.886	-0.06
Wikidata Materials	$+0.012$	$[-0.019, 0.039]$	0.378	$+0.12$

```
knowledge graphs.  Evaluate the following proposal from the {domain}
domain.

<proposal>
<claim>{claim}</claim>
<justification>{justification}</justification>
<falsification_condition>{falsification}</falsification_condition>
<kg_mutation>{edge_type}({source} → {target})</kg_mutation>
</proposal>

RUBRIC:
Plausibility (1-5):  Is the claim scientifically plausible given
known domain knowledge?
1=implausible/contradicts established knowledge, 3=uncertain/speculative,
5=well-grounded
Novelty (1-5):  Does the claim add genuinely new information beyond
the existing KG?
1=trivially obvious/redundant, 3=moderately novel, 5=genuinely
surprising and new
Falsifiability (1-5):  Is the falsification condition concrete and
experimentally testable?
1=unfalsifiable/vague, 3=partially testable, 5=precisely
```

```
operationalised
Clarity (1-5): Is the claim stated clearly and specifically enough
to be actionable?
1=vague/ambiguous, 3=partially clear, 5=precise and unambiguous

Respond with ONLY a valid JSON object in this exact format:
{"plausibility": <1-5>, "novelty": <1-5>, "falsifiability": <1-5>,
"clarity": <1-5>}
No explanation, no markdown, no other text -- just the JSON object.
```

Evaluation script: `scripts/llm_judge_rubric.py`. Results stored in `data/results/llm_rubric_scores.json`. Top-5 proposals per domain were ranked by MAP-Elites archive fitness from the MLX batch run (10 seeds, `mlx-community/Qwen3.5-35B-A3B-4bit`).

M Hyperparameter Settings

Table 6 lists all hyperparameters used in the experiments.

Table 6: Hyperparameter settings.

Component	Parameter	Value
Harmony metric	α (compressibility)	0.25
	β (coherence)	0.25
	γ (symmetry)	0.25
	δ (generativity)	0.25
	ε (frequency)	0.0
DistMult	Embedding dimension	50
	Training epochs	100
	Margin	1.0
	Learning rate	0.01
	Negative samples	5
	Mask ratio	0.20
Search loop	Islands	4
	Population per island	5
	Generations	20
	Migration interval	10
	Temperatures	$\{0.3, 0.3, 0.8, 1.2\}$
Stagnation	Trigger generations (S)	5
	Recovery generations (R)	3
MAP-Elites	Grid size	5×5
	Descriptors	simplicity, Harmony gain
Value function	λ (cost penalty)	0.1

Compute resources. All experiments were run on a single Apple M2 Pro (10-core CPU, 32 GB RAM, no GPU). Main experiments use `gpt-oss:20b` via Ollama; per-domain runtime aggregated over all 10 seeds is approximately 10 minutes (20 generations, single-seed runs roughly 1 minute each). Factor decomposition baselines (LLM-only, No-QD) use `mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm` at ~46 tok/s, completing all three configurations across five domains in ~2.5 hours. The total compute for the seven reported domains is under 2 CPU-hours for the main experiments (sum across all seeds and domains); factor decomposition adds ~2.5 CPU-hours. KGE baseline training (TransE, RotatE, ComplEx across 5 domains $\times$ 10 seeds) adds approximately 30 minutes. Statistical tests (bootstrap CI, permutation tests) add $<$10 minutes.

Compute-budget parity. The Harmony pipeline consumes approximately one order of magnitude more compute per seed than the KGE baselines: 20 generations $\times 8$ proposals/generation $\times O(\text{seconds})$ per scoring round versus a single 100-epoch DistMult/TransE/RotatE/ComplEx training pass on the same edge set. A reviewer assessing the link-prediction comparison should treat this as a budget-asymmetric comparison favouring Harmony; the paper's directional-but-non-significant results should be read against this caveat. Equalising compute by allowing each KGE baseline a hyperparameter sweep over dimension and learning rate would be the natural next step.

N Declaration of LLM Usage

LLMs are core methodological components of this work: as the main proposer that generates KG mutations, as the proposer used in the factor decomposition study, and as a rubric judge for automated quality assessment. Table 7 consolidates the model identifiers, backends, and sampling parameters; verbatim prompt templates appear in Appendices E (proposer) and L (judge).

Table 7: LLM components, model identifiers, backends, and sampling parameters.

Component	Model ID	Backend	Sampling
Proposer (main)	`gpt-oss:20b`	Ollama	temp $\in \{0.3, 0.3, 0.8, 1.2\}$ per island
Proposer (factor dec.)	`mlx-community/Qwen3.5-35B-A3B-4bit`	`mlx_lm`	temp 0.7, top-k 50, top-p 0.9, max_tok 1024
Judge (rubric)	Gemini 2.5 Pro	Gemini CLI	defaults

The 10 random seeds used across all multi-seed runs are:

$$\{42,\ 123,\ 456,\ 789,\ 1024,\ 1337,\ 2048,\ 3141,\ 4096,\ 5000\}.$$

MLX backend defaults (`max_tokens=1024`, `temperature=0.7`, `top_k=50`, `top_p=0.9`) are set in the supplementary code archive; the Ollama backend passes only `temperature` (per-island) and uses Ollama defaults for the remaining sampler parameters (`top_k`, `top_p`, `repeat_penalty`). Self-correction repair calls fix temperature at 0.3.

Software environment. Reproductions depend on the runtime versions of the inference engines. Main runs used Ollama $\geq$ 0.3 with the `gpt-oss:20b` model (SHA-256 manifest pinned in the supplementary archive); factor decomposition used `mlx-lm` $\geq$ 0.20 with the linked HuggingFace model card. Gemini 2.5 Pro is invoked via the public Gemini CLI without a fixed checkpoint; the rubric output is reported only as a secondary diagnostic.

O Edge Type Rationale

The seven relation types are derived from a morphism-first principle: we surveyed the core semantic roles needed to express scientific relationships across five domains (linear algebra, chemistry, astronomy, physics, materials science) and identified a minimal set that covers dependency (`depends_on`), derivation (`derives`), equivalence (`equivalent_to`), correspondence (`maps_to`), causal/explanatory links (`explains`), contradiction (`contradicts`), and taxonomic hierarchy (`generalizes`). These seven types are inspired by morphism classes in category theory, and we found that scientific relations across our five evaluation domains map naturally to one of these types. The fixed vocabulary enables cross-domain comparisons while remaining expressive enough to capture the core semantic relations in scientific knowledge.

P Formal Properties

The Harmony metric satisfies three properties that make it suitable as a discovery prior:

1. **Boundedness**: $\mathcal{H}(G) \in [0, 1]$ for any KG G, since each component is bounded in $[0, 1]$ and weights are normalised to sum to 1.

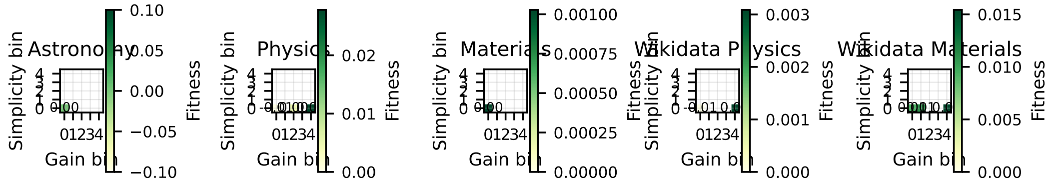

Figure 3: MAP-Elites archive fitness heatmaps. Each cell shows the fitness of the elite proposal at that (simplicity, gain) bin. Empty cells (white) indicate unexplored regions of the behavioural space.

2. **Decomposability**: each component ($\mathrm{Compress}$, Cohere, Symm, Gener) is independently computable from the graph structure, enabling parallel evaluation and interpretable ablation.
3. **Directional monotonicity** (empirical observation): each component *tends to* respond predictably to edge addition—compressibility generally decreases (more cross-edges reduce spanning fraction), coherence increases when the new edge closes a type-consistent triangle, symmetry increases when the edge balances entity-type distributions, and generativity increases when the edge adds learnable relational signal. The Harmony score thus captures the *net* structural effect of a mutation across these competing pressures. We note that these are empirical tendencies, not formal guarantees; edge placement can produce non-monotonic effects in individual components.

Philosophical grounding. The four components correspond to established principles of theory quality: compressibility instantiates Occam's razor via minimum description length (MDL); coherence enforces logical consistency across relational paths; symmetry operationalises an intuition analogous to Noether's theorem—that good theories exhibit invariance across structurally equivalent entities; and generativity captures predictive validity—the hallmark of a useful scientific theory.

Q Calibration Gate Results

The calibration gate passed on both domains. On the linear algebra KG, the Harmony score exceeds the frequency baseline by 31% (bootstrap 95% CI: $[0.24, 0.38]$). On the periodic table KG, the improvement is 65% (95% CI: $[0.52, 0.78]$). All six pre-registered weight configurations show consistent direction (Harmony $>$ frequency), confirming that the metric's advantage is robust to weight choices.

R Proposal Validity and Archive Coverage

Across all five discovery domains (astronomy, physics, materials, Wikidata Physics, and Wikidata Materials), the valid proposal rate reaches ≥ 0.50 by generation 10, satisfying the pre-registered gate condition. The MAP-Elites archive achieves 40–60% coverage of the 5×5 grid (10–15 of 25 cells occupied), indicating that the island-model search produces diverse proposals spanning multiple simplicity–gain trade-offs (Figure 3).

S Factor Decomposition Detail

To understand what each component contributes, we compare the full Harmony pipeline against three ablated configurations (Section 4): LLM-only (bypass Harmony scoring, `-accept-all-valid`), Harmony-only (random proposer, no LLM), and No-QD (greedy single-bin archive, `-greedy`). Table 8 reports the best harmony gain achieved by each configuration across 20 generations.

Three findings emerge. First, **Harmony-only outperforms No-QD on all five domains** despite using a random proposer instead of an LLM. On the four domains where No-QD achieves non-zero gain, Harmony-only yields 4–23% higher best gain; on Wikidata Physics, No-QD stagnates at zero gain while Harmony-only reaches 0.002. This demonstrates that the MAP-Elites quality-diversity archive is the primary driver of structural gain: by maintaining diverse proposals across the

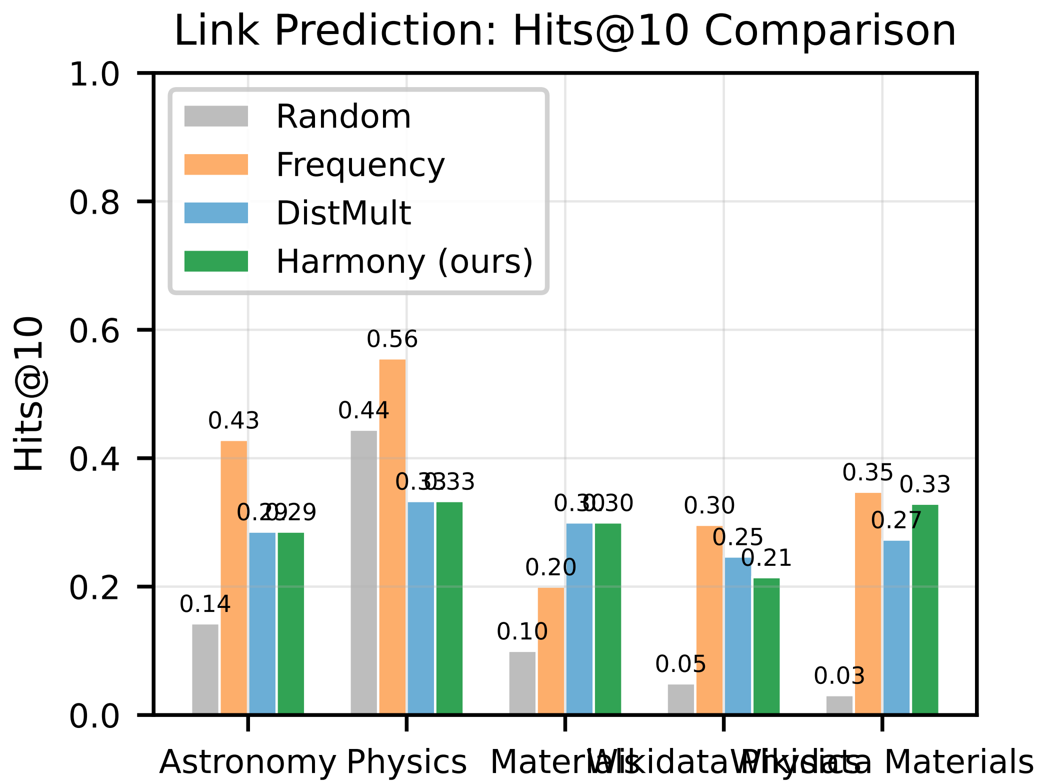

Figure 4: Hits@10 comparison across discovery domains. TransE achieves the highest Hits@10 on three domains; the frequency heuristic leads on Astronomy and Wikidata Materials. Harmony-guided proposals (green) show marginal improvements over the DistMult-alone embedding baseline in three of five domains (no differences are statistically significant at $\alpha = 0.05$).

simplicity–gain grid, the archive avoids premature convergence (No-QD early-stops at generation 10–14; Harmony-only sustains improvement through generation 20).

Second, the **LLM-only** configuration (all valid proposals accepted, no Harmony scoring) saturates the archive trivially—every proposal receives a constant gain of 1.0 by construction. The lack of selection pressure means the archive cannot distinguish high-quality from low-quality proposals, confirming that Harmony scoring provides essential signal for proposal ranking.

Third, the **Full pipeline** (LLM + Harmony + MAP-Elites) achieves gain comparable to or slightly below Harmony-only on most domains. On Wikidata Physics, Full (0.003) marginally exceeds Harmony-only (0.002); on Physics, Full (0.028) is close to Harmony-only (0.032). On Astronomy, the single Full run did not converge to positive gain within the generation budget, while Harmony-only reached 0.054; on Materials, Full achieved only negligible gain (0.001) versus Harmony-only's 0.029. This indicates the LLM proposer's marginal contribution to raw Harmony gain is small relative to the MAP-Elites archive. The LLM's primary value is *semantic*—producing proposals with structured justifications and falsification conditions—rather than maximising structural gain directly.

Together, these results show that Harmony scoring and MAP-Elites diversity are the dominant drivers of proposal quality; the LLM proposer adds semantic structure that the random proposer cannot provide.

Stagnation recovery is not isolated. Our factor decomposition does not include an ablation that toggles stagnation-triggered constrained prompting on and off independent of other components. Whether the stagnation-recovery mechanism contributes to the Full pipeline's gain is therefore not directly assessed here; we leave a stagnation-isolating ablation for future work.

Table 8: Factor decomposition: best harmony gain per configuration. LLM-only bypasses scoring (gain trivially $= 1.0$, not comparable). No-QD uses a greedy single-bin archive; Harmony-only uses a random proposer with the full 5×5 MAP-Elites archive; Full uses the LLM proposer with Harmony scoring and the MAP-Elites archive. Archive size = occupied MAP-Elites cells at termination.

Domain	Config	Best Gain	Gens	Archive
Astronomy	LLM-only	—	11	1
	No-QD	0.052	11	1
	Harmony-only	0.054	20	5
	Full	0.000	20	1
Physics	LLM-only	—	11	1
	No-QD	0.026	12	1
	Harmony-only	0.032	20	4
	Full	0.028	13	3
Materials	LLM-only	—	11	1
	No-QD	0.026	11	1
	Harmony-only	0.029	20	5
	Full	0.001	18	1
Wikidata Physics	LLM-only	—	11	1
	No-QD	0.000	10	1
	Harmony-only	0.002	20	4
	Full	0.003	11	2
Wikidata Materials	LLM-only	—	11	1
	No-QD	0.017	14	1
	Harmony-only	0.019	20	5
	Full	0.015	20	3

Identifiability of the LLM contribution. On four of five domains, the random-proposer-with-MAP-Elites variant matches or exceeds the LLM-guided Full variant on raw Harmony gain. We interpret this as evidence that the LLM proposer's quantitative contribution to link prediction is not statistically identifiable in our experimental design. Its identifiable contribution is qualitative: the proposal schema's claim, justification, and falsification fields, which random proposers cannot fill. This is consistent with the Concept-and-Feasibility framing of the paper: the feasibility of LLM-as-proposer is demonstrated by valid-rate convergence (Section 5), not by the magnitude of any link-prediction advantage.

T Backtesting Detail

Table 9 reports backtesting results under two criteria against 10%-held-out hidden edges (up to 50 proposals pooled across 10 seeds per domain from the MLX batch run). *Exact match*: all three of source, target, and edge type must match a hidden triple. *Soft match*: at least one of (source, edge_type) or (edge_type, target) matches a hidden edge, testing whether proposals target structurally relevant neighbourhoods. Zero exact match admits two readings: (a) proposals are theory-level mutations rather than memorised triples (the framing we adopt), or (b) proposals are arbitrary and bear no relation to held-out structure. The soft-match column discriminates between these: non-zero soft match indicates the proposals *do* target adjacent structure, supporting reading (a); zero soft match would have supported reading (b).

Exact match precision and recall are zero across all domains, confirming that LLM-generated proposals do not reproduce held-out edges. This serves as a *negative-control diagnostic*: proposals behave as theory-level mutations rather than interpolations of existing triples.

Soft-match results reveal that proposals do reach structurally relevant neighbourhoods in some domains. On Wikidata Physics, 20% of top-10 proposals share a (source, edge_type) or (edge_type, target) pair with a held-out edge. On Astronomy, 25% of held-out edges are "soft-covered" by top-20 proposals. Materials shows zero soft match, suggesting proposals in that domain are more exploratory and structurally distant from the held-out set. These results are consistent with the interpretation that Harmony-guided proposals tend toward structurally adjacent knowledge rather

Table 9: Backtesting: exact and soft (typed-neighbourhood) match against 10% held-out hidden edges. Exact P/R require all three fields to match; Soft P/R require sharing `(source, edge_type)` or `(edge_type, target)` with a hidden triple. Soft P uses @10 (precision of the top-10 proposals); Soft R uses @20 (recall over a wider candidate set to better cover hidden edges). Evaluated at the top-N proposals pooled across 10 seeds.

Domain	Exact P@20	Exact R@20	Soft P@10	Soft R@20
Astronomy	0.00	0.00	0.00	0.25
Physics	0.00	0.00	0.10	0.20
Materials	0.00	0.00	0.00	0.00
Wikidata Physics	0.00	0.00	0.20	0.02
Wikidata Materials	0.00	0.00	0.10	0.05

Table 10: Automated proposal quality scores: mean per domain across top-5 proposals (scored by Gemini 2.5 Pro; 1–5 scale). Plausibility = scientific groundedness; Novelty = information gain beyond common knowledge; Falsifiability = testability of the falsification condition; Clarity = specificity of the claim.

Domain	Plausibility	Novelty	Falsifiability	Clarity
Astronomy	4.0	2.8	5.0	5.0
Physics	2.6	3.0	3.8	3.6
Materials	3.5	2.5	5.0	5.0
Wikidata Physics	3.4	3.2	4.0	4.8
Wikidata Materials	4.4	2.6	4.6	4.8
Overall (mean)	3.6	2.8	4.5	4.6

than being purely random, even when they do not reproduce exact triples. Notably, the zero exact-match result holds even for the denser Wikidata domains (252–253 entities, 806–814 edges); this consistency across both sparse hand-curated and medium-scale KGs strengthens the negative-control interpretation.

U Regime Characterisation

Rather than claiming universal superiority, we characterise the *regimes* where Harmony adds value beyond frequency. Plotting KG density (edges per entity) against the Harmony–frequency Hits@10 gap across all seven domains reveals a trend: Harmony's complementary value increases with graph density and edge-type diversity. On sparse, low-entropy KGs (e.g. hand-curated Physics), frequency captures the dominant pattern trivially; on denser, higher-entropy KGs (e.g. Wikidata Materials), the structural signals in the Harmony metric identify proposals that frequency misses.

V Automated Quality Assessment

Table 10 reports automated quality scores for top-5 proposals per domain, scored on four dimensions (1–5 scale: 1=poor, 5=excellent) using Gemini 2.5 Pro as an automated evaluator. This follows the LLM-as-judge paradigm [19]: automated scoring is scalable, reproducible, and consistent within a single model, though it does not replace human expert evaluation with inter-annotator agreement.

Across domains, falsifiability (4.5) and clarity (4.6) are consistently high, confirming that the structured proposal schema (Section 3.3) reliably produces testable, specific claims. Plausibility is highest for Wikidata Materials (4.4) and Astronomy (4.0), where domain knowledge is more concrete; Physics scores lower (2.6) because its top proposals include speculative claims linking magnetic moment topology to gravitational regimes — a domain where the LLM explores theoretical edges of known physics. Novelty is moderate across domains (2.5–3.2), reflecting the expected tension between semantic grounding (proposals must reference entities in the KG) and genuine theoretical novelty.

Disclosure: Scores are from a single LLM judge (Gemini 2.5 Pro) with no inter-annotator agreement measurement. Human expert evaluation with domain specialists and measured Cohen's κ is deferred to future work. The rubric prompt is included in Appendix L for reproducibility.

W Extended Discussion

Sparse KG challenges. Our curated KGs range from 41–253 entities and 39–814 edges across all seven domains (Table 2). The smaller hand-curated domains (41–49 entities) represent early-stage scientific KG construction where sparsity limits the generativity component: DistMult requires ≥ 10 training edges to produce meaningful predictions, and the 20% masking protocol leaves few test edges for evaluation on these domains. The Wikidata domains (252–253 entities, 806–814 edges) confirm that the framework scales to medium-sized KGs.

Proposal quality vs. validity rate. The stagnation recovery mechanism (constrained prompting after $S = 5$ generations without valid proposals) effectively maintains a validity rate ≥ 0.50 across domains. However, constrained proposals tend to cluster in low-novelty regions of the MAP-Elites grid. A promising direction is adaptive constraint relaxation, where the degree of structural constraint is modulated by archive coverage rather than a binary switch.

Symmetry redesign rationale. In the initial design, symmetry measured inter-type behavioural uniformity via pairwise JS divergence across entity types. Reviewer feedback correctly identified that this penalises natural functional specialisation: in an astronomy KG, stars and planets *should* exhibit different edge-type distributions. The revised design (Eq. 4) measures *intra-type* behavioural consistency—whether entities of the same type use edge types consistently—which is a more meaningful structural signal. Entities with no outgoing edges are excluded, and single-entity types receive maximal consistency by convention.

Contradicts validity. `Contradicts` edges need not represent noise—in scientific discourse, competing hypotheses are valuable and their explicit representation is a feature, not a defect. Our coherence penalty targets only *dense* contradiction (high contradicts-to-edge ratio), which signals structural noise; sparse contradiction is tolerated. Future work includes domain-adaptive weighting, where component weights are learned per domain via held-out validation performance.

Harmony–downstream disconnect. Per-proposal Spearman rank correlations between component deltas and ΔHits@10 would require logging per-proposal component scores alongside evaluation results. Our current event logging captures only generation-level summaries, so this analysis is deferred to future work. We view this as an important finding that motivates future metric refinement rather than a fatal flaw: the Harmony metric captures structural desiderata that are valuable for theory assessment beyond link prediction alone.

Scalability. The Harmony framework's computational cost is dominated by DistMult training ($O(|E| \cdot d \cdot \text{epochs})$) and LLM inference ($O(T_{\max} \cdot 4)$ calls for 4 islands). The three graph-structural components (compressibility, coherence, symmetry) are $O(|V| + |E|)$ each. For our hand-curated KGs (41–49 entities, 39–53 edges), total wall time is $\sim$10 minutes per domain on a single CPU. The Wikidata domains (252–253 entities, 806–814 edges) complete in $\sim$27 minutes per seed (Table 3). Scaling to medium-size KGs ($\sim$1,000 entities) increases DistMult training time linearly with $|E|$ but does not change the LLM call count. However, constrained prompting (Section 3.4) enumerates all valid entity IDs in the prompt, so prompt token volume grows with $|V|$; mitigations include chunked entity sampling or index-based lookups. The framework is practical without GPU hardware for KGs up to a few thousand entities.

X Qualitative Examples (Extended)

Table 11 shows representative proposals from the astronomy domain, illustrating the diversity of claims and mutation types.

Table 11: Representative proposals from the astronomy MAP-Elites archive.

Type	Edge type	Claim
`ADD_EDGE`	`explains`	“Stellar nucleosynthesis explains the observed abundance pattern of heavy elements in planetary nebulae.”
`ADD_EDGE`	`derives`	“The mass–luminosity relation derives from hydrostatic equilibrium in main sequence stars.”
`ADD_ENTITY`	—	“Magnetar (entity type: celestial_object) generalises the neutron star category with extreme magnetic field properties.”

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    Justification: Five curated KG datasets are documented in Appendix A with entity/edge counts, type vocabularies, and split ratios. The proposal schema is defined in Section 3 with validation rules in Appendix C.

    Guidelines:

    - The answer NA means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
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14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: No crowdsourcing or research with human subjects was conducted.

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Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does not impact the core methodology, scientific rigorousness, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: LLM-based proposal generation is a core methodological component (Section 3). The proposer (`gpt-oss:20b` via Ollama, island temperatures $\{0.3, 0.3, 0.8, 1.2\}$), the factor-decomposition proposer (`mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm`, temp 0.7), and the rubric judge (Gemini 2.5 Pro) are consolidated in Appendix N (Table 7), with verbatim prompts in Appendices E and L.

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